

Proposition 5. *Suppose $K \geq 2$ is even. Then, for $R > 1$ sufficiently large,*

$$d_{\min}^2 \geq 8E\eta. \quad (\star)$$

Before proceeding with the proof, we recall the inner product formula [1, Eq. (13)]

$$I_{\mathbf{s}}(R) = \frac{1}{K+1} \sum_{\ell=0}^K A_{\mathbf{H}}(\zeta^\ell) P_{\mathbf{s}}(\zeta^\ell), \quad \zeta = e^{j2\pi/(K+1)}, \quad (1)$$

where

$$A_{\mathbf{H}}(\zeta^\ell) = E \left[1 - 2\eta \cos \left(2\pi \frac{\ell}{K+1} \right) \right], \quad \eta = \frac{1}{R^K + R^{-K}}, \quad (2)$$

$$P_{\mathbf{s}}(\zeta^\ell) = \prod_{k=0}^{K-1} B_k(\zeta^\ell)^{s_k}, \quad \mathbf{s} \in \{-1, 0, 1\}^K, \quad (3)$$

$$B_k(\zeta^\ell) = R^{-1} \frac{\zeta^\ell - R\phi^k}{\zeta^\ell - R^{-1}\phi^k}, \quad \phi = e^{j2\pi/K}. \quad (4)$$

Now for the proof.

Proof. Our approach is as follows:

- 1) develop a tight approximation for $|I_{\mathbf{s}}(R)|$ when $R > 1$ is large;
- 2) show that $\pm \mathbf{s}_a = \pm[1, -1, \dots, 1, -1]^T$ maximize this approximation for even $K \geq 2$;
- 3) simplify $2E - 2|I_{\mathbf{s}_a}(R)|$ to obtain the lower bound in $(\star)$.

Step 1) Let us assume a unit energy¹ $E = 1$ and expand $|I_{\mathbf{s}}(R)|$ as

$$|I_{\mathbf{s}}(R)| = \frac{1}{K+1} \left| \sum_{\ell=0}^K A_{\mathbf{H}}(\zeta^\ell) P_{\mathbf{s}}(\zeta^\ell) \right| = \frac{1}{K+1} \left| \sum_{\ell=0}^K P_{\mathbf{s}}(\zeta^\ell) - 2\eta \sum_{\ell=0}^K \cos \left(2\pi \frac{\ell}{K+1} \right) P_{\mathbf{s}}(\zeta^\ell) \right|.$$

Since $|B_k(\zeta^\ell)| = 1$ for all $\ell \in [K+1]$ and $k \in [K]$, $|P_{\mathbf{s}}(\zeta^\ell)| = 1$. Also, $\eta < R^{-K}$ for all $R > 1$, and $\cos(\cdot)$ is bounded. Thus

$$|I_{\mathbf{s}}(R)| = \frac{1}{K+1} \left| \sum_{\ell=0}^K P_{\mathbf{s}}(\zeta^\ell) \right| + O(R^{-K}), \quad (5)$$

so the large- R behavior of $|I_{\mathbf{s}}(R)|$ is governed by $P_{\mathbf{s}}(\zeta^\ell)$ [up to a bounded error $O(R^{-K})$]. Define

$$P_{\mathbf{s}}^{(\infty)}(\zeta^\ell) := \lim_{R \rightarrow \infty} P_{\mathbf{s}}(\zeta^\ell).$$

¹The energy $E > 0$ simply scales $|I_{\mathbf{s}}(R)|$, so we set $E = 1$ (without loss of generality).

Noting that $\lim_{R \rightarrow \infty} B_k(\zeta^\ell) = -\phi^k \zeta^{-\ell}$ [see (4)], $P_{\mathbf{s}}^{(\infty)}(\zeta^\ell)$ simplifies as

$$\begin{aligned}
P_{\mathbf{s}}^{(\infty)}(\zeta^\ell) &= \prod_{k: s_k=1} (-\phi^k \zeta^{-\ell}) \prod_{k: s_k=-1} (-\phi^*)^k \zeta^\ell \\
&= (-1)^{v(\mathbf{s})} \zeta^{\ell u(\mathbf{s})} \prod_{k: s_k=1} (\phi^k) \prod_{k: s_k=-1} (\phi^*)^k \\
&= (-1)^{v(\mathbf{s})} \zeta^{\ell u(\mathbf{s})} \exp \left(j \frac{2\pi}{K} \left(\sum_{k: s_k=1} k - \sum_{k: s_k=-1} k \right) \right) \\
&= (-1)^{v(\mathbf{s})} \zeta^{\ell u(\mathbf{s})} \exp \left(j \frac{2\pi}{K} \sum_{k=0}^{K-1} k s_k \right), \tag{6}
\end{aligned}$$

where $u(\mathbf{s}) := \sum_{k=0}^{K-1} s_k$ and $v(\mathbf{s}) := \sum_{k=0}^{K-1} s_k^2$. In light of (5), this implies

$$\lim_{R \rightarrow \infty} |I_{\mathbf{s}}(R)| = \frac{1}{K+1} \left| \sum_{\ell=0}^K P_{\mathbf{s}}^{(\infty)}(\zeta^\ell) \right| = \frac{1}{K+1} \left| \sum_{\ell=0}^K \zeta^{\ell u(\mathbf{s})} \right| = \delta_{u(\mathbf{s})0},$$

since $\sum_{\ell=0}^K = (K+1)\delta_{u(\mathbf{s})0}$. Hence balanced patterns (i.e., $\mathbf{s} \in \{-1, 0, 1\}^K$ with $u(\mathbf{s}) = 0$) are the only candidate maximizers of $|I_{\mathbf{s}}(R)|$ for large R .

With this in mind, we factor $-\phi^k \zeta^{-\ell}$ out of $B_k(\zeta^\ell)$ as

$$B_k(\zeta^\ell) = -\phi^k \zeta^{-\ell} G_k(\zeta^\ell), \quad G_k(\zeta^\ell) := \frac{1 - R^{-1}(\phi^*)^k \zeta^\ell}{1 - R^{-1}\phi^k \zeta^{-\ell}}.$$

Then $P_{\mathbf{s}}(\zeta^\ell) = P_{\mathbf{s}}^{(\infty)}(\zeta^\ell) \prod_{k=0}^{K-1} G_k(\zeta^\ell)^{s_k}$, and for balanced $\mathbf{s} \in \{-1, 0, 1\}^K$, $P_{\mathbf{s}}^{(\infty)}(\zeta^\ell) = P_{\mathbf{s}}^{(\infty)}$ is independent of ℓ with $|P_{\mathbf{s}}^{(\infty)}| = 1$ [see (6)]. Thus if $u(\mathbf{s}) = 0$, (5) further simplifies to

$$\begin{aligned}
|I_{\mathbf{s}}(R)| &= \frac{1}{K+1} \left| \sum_{\ell=0}^K P_{\mathbf{s}}^{(\infty)} \prod_{k=0}^{K-1} G_k(\zeta^\ell)^{s_k} \right| + O(R^{-K}) \\
&= \frac{1}{K+1} \left| \sum_{\ell=0}^K \prod_{k=0}^{K-1} G_k(\zeta^\ell)^{s_k} \right| + O(R^{-K}). \tag{7}
\end{aligned}$$

Towards approximating the sum in (7), we compute²

$$\text{Log} G_k(\zeta^\ell) = \text{Log}(1 - R^{-1}(\phi^*)^k \zeta^\ell) - \text{Log}(1 - R^{-1}\phi^k \zeta^{-\ell}), \tag{8}$$

where for any $R > 1$, $|R^{-1}(\phi^*)^k \zeta^\ell| < 1$ (similarly for $R^{-1}\phi^k \zeta^{-\ell}$). Invoking the power series $\text{Log}(1 - z) = -\sum_{m \geq 1} z^m/m$, converging absolutely for $|z| < 1$, (8) becomes

$$\begin{aligned}
\text{Log} G_k(\zeta^\ell) &= -\sum_{m \geq 1} \frac{R^{-m}}{m} (\phi^*)^{km} \zeta^{\ell m} + \sum_{m \geq 1} \frac{R^{-m}}{m} \phi^{km} \zeta^{-\ell m} \\
&= \sum_{m \geq 1} \frac{R^{-m}}{m} (\phi^{km} \zeta^{-\ell m} - (\phi^*)^{km} \zeta^{\ell m}). \tag{9}
\end{aligned}$$

²See Appendix A for a justification of the identity $\text{Log}(z_1/z_2) = \text{Log}(z_1) - \text{Log}(z_2)$ applied here.

Hence³

$$\begin{aligned}
\text{Log} \prod_{k=0}^{K-1} G_k(\zeta^\ell)^{s_k} &= \sum_{k=0}^{K-1} s_k \text{Log} G_k(\zeta^\ell) \\
&= \sum_{k=0}^{K-1} s_k \sum_{m \geq 1} \frac{R^{-m}}{m} (\phi^{km} \zeta^{-\ell m} - (\phi^*)^{km} \zeta^{\ell m}) \\
&\stackrel{(a)}{=} \sum_{m \geq 1} \frac{R^{-m}}{m} \left(\zeta^{-\ell m} \sum_{k=0}^{K-1} s_k \phi^{km} - \zeta^{\ell m} \sum_{k=0}^{K-1} s_k (\phi^*)^{km} \right) \\
&= \sum_{m \geq 1} \frac{R^{-m}}{m} (\zeta^{-\ell m} F_m(\mathbf{s}) - \zeta^{\ell m} F_m^*(\mathbf{s})), \tag{10}
\end{aligned}$$

where $F_m(\mathbf{s}) := \sum_{k=0}^{K-1} s_k \phi^{km}$. The exchange of sums in (a) is justified as the double sum converges absolutely ($\sum_{k=0}^{K-1} s_k$ is finite and (9) converges absolutely; for example, see [here](#)). Exponentiating both sides of (10), we get

$$\begin{aligned}
\prod_{k=0}^{K-1} G_k(\zeta^\ell)^{s_k} &= \exp \left(\sum_{m \geq 1} \frac{R^{-m}}{m} (\zeta^{-\ell m} F_m(\mathbf{s}) - \zeta^{\ell m} F_m^*(\mathbf{s})) \right) \\
&= \exp(T_\ell(\mathbf{s})) \\
&\stackrel{(b)}{=} 1 + T_\ell(\mathbf{s}) + \frac{1}{2} T_\ell(\mathbf{s})^2 + O(T_\ell(\mathbf{s})^3), \tag{11}
\end{aligned}$$

where $T_\ell(\mathbf{s}) := \sum_{m \geq 1} \frac{R^{-m}}{m} (\zeta^{-\ell m} F_m(\mathbf{s}) - \zeta^{\ell m} F_m^*(\mathbf{s}))$ and the identity in (b) follows from the Taylor expansion of $\exp(\cdot)$.

We can now obtain an approximation of the sum in (7) by summing (11) over $\ell \in [K+1]$. Summing the constant term yields $\sum_{\ell=0}^K 1 = K+1$, while

$$\sum_{\ell=0}^K T_\ell(\mathbf{s}) = \sum_{m \geq 1} \frac{R^{-m}}{m} \left(F_m(\mathbf{s}) \sum_{\ell=0}^K \zeta^{-\ell m} - F_m^*(\mathbf{s}) \sum_{\ell=0}^K \zeta^{\ell m} \right) = O(R^{-(K+1)}),$$

since $\sum_{\ell=0}^K \zeta^{\pm \ell m} = 0$ unless $m \equiv 0 \pmod{K+1}$. To evaluate the third sum $\frac{1}{2} \sum_{\ell=0}^K T_\ell(\mathbf{s})^2$, it is useful to define $m_0 := \min\{m \geq 1 : F_m(\mathbf{s}) \neq 0\}$. A fact we use later is that $m_0 \leq K/2$.⁴ Here, however, we note that $F_m(\mathbf{s}) = 0$ for all $m < m_0$ implies

$$\begin{aligned}
T_\ell(\mathbf{s}) &= \sum_{m \geq m_0} \frac{R^{-m}}{m} (\zeta^{-\ell m} F_m(\mathbf{s}) - \zeta^{\ell m} F_m^*(\mathbf{s})) \\
&= \frac{R^{-m_0}}{m_0} (\zeta^{-\ell m_0} F_{m_0}(\mathbf{s}) - \zeta^{\ell m_0} F_{m_0}^*(\mathbf{s})) + O(R^{-(m_0+1)}).
\end{aligned}$$

³See Appendix A for a justification of the identity $\text{Log}(\prod_k z_k^{s_k}) = \sum_k s_k \text{Log}(z_k)$ applied here.

⁴ $F_m(\mathbf{s})$ is the m th entry of the DFT of $\mathbf{s} \in \mathbb{R}^K$, so Hermitian symmetry forces $m_0 \leq K/2$.

Thus

$$\begin{aligned}
\frac{1}{2} \sum_{\ell=0}^K T_{\ell}(\mathbf{s})^2 &= \frac{1}{2} \sum_{\ell=0}^K \frac{R^{-2m_0}}{m_0^2} \left(\zeta^{-2\ell m_0} F_{m_0}(\mathbf{s})^2 - \zeta^{2\ell m_0} F_{m_0}^*(\mathbf{s})^2 - 2|F_{m_0}(\mathbf{s})|^2 \right) + O(R^{-(2m_0+1)}) \\
&= \frac{1}{2} \frac{R^{-2m_0}}{m_0^2} \left(F_{m_0}(\mathbf{s})^2 \sum_{\ell=0}^K \zeta^{-2\ell m_0} - F_{m_0}^*(\mathbf{s})^2 \sum_{\ell=0}^K \zeta^{2\ell m_0} \right. \\
&\quad \left. - 2 \sum_{\ell=0}^K |F_{m_0}(\mathbf{s})|^2 \right) + O(R^{-(2m_0+1)}) \\
&\stackrel{(c)}{=} -(K+1) \frac{|F_{m_0}(\mathbf{s})|^2}{m_0^2} R^{-2m_0} + O(R^{-(2m_0+1)}),
\end{aligned}$$

where (c) follows since $1 \leq m_0 \leq K/2$ implies $2m_0 \not\equiv 0 \pmod{K+1}$ and $\sum_{\ell=0}^K \zeta^{\pm 2\ell m_0} = 0$. Combining these results and noting that $T_{\ell}(\mathbf{s})^3 = O(R^{-3m_0})$, $3m_0 \geq 2m_0 + 1$, (7) becomes

$$|I_{\mathbf{s}}(R)| = \left| 1 - \frac{|F_{m_0}(\mathbf{s})|^2}{m_0^2} R^{-2m_0} \right| + O(R^{-(2m_0+1)}) + O(R^{-K}). \quad (12)$$

Step 2) For $m_0 < K/2$, $2m_0 < K$ and $R^{-(2m_0+1)} = o(R^{-2m_0})$, so (12) strengthens to

$$|I_{\mathbf{s}}(R)| = \left| 1 - \frac{|F_{m_0}(\mathbf{s})|^2}{m_0^2} R^{-2m_0} \right| + o(R^{-2m_0}).$$

For $m_0 = K/2$, however, $2m_0 = K$ such that $|I_{\mathbf{s}}(R)| = 1 + O(R^{-K})$, which is equivalent to

$$1 - |I_{\mathbf{s}}(R)| = O(R^{-K}).$$

But $R^{-K} = o(R^{-2m_0})$ for each $m_0 < K/2$. This implies that, for sufficiently large $R > 1$, any signed pattern with $m_0 = K/2$ has a larger $|I_{\mathbf{s}}(R)| \leq 1$ than every signed pattern with $m_0 < K/2$.

We now identify the signed patterns satisfying $m_0 = K/2$. It is clear that such patterns exist if and only if K is even. Hence suppose K is even and that $m_0 = K/2$. By definition of m_0 , $F_0(\mathbf{s}) = \dots = F_{K/2-1}(\mathbf{s}) = 0$, so Hermitian symmetry implies $F_m(\mathbf{s}) = 0$ for all $m \neq K/2$.⁵ An IDFT then yields

$$s_k = \frac{1}{K} \sum_{m=0}^{K-1} F_m(\mathbf{s}) \phi^{-mk} = \frac{F_{K/2}(\mathbf{s})}{K} \phi^{-\frac{K}{2}k} = \frac{F_{K/2}(\mathbf{s})}{K} (-1)^k \in \{-1, 0, 1\}, \quad k \in [K].$$

It follows that $F_{K/2}(\mathbf{s})/K = \pm 1$, meaning the only signed patterns with $m_0 = K/2$ are the alternating patterns $\pm \mathbf{s}_a = \pm[1, -1, \dots, 1, -1]^T$.

⁵Recall that $F_m(\mathbf{s})$ is the m th entry of the DFT of $\mathbf{s} \in \{-1, 0, 1\}^K$, where we exclude $\mathbf{s} = [0, \dots, 0]^T$ (which corresponds to a self distance).

Step 3) Plugging the alternating pattern $\mathbf{s}_a = [1, -1, \dots, 1, -1]^T$ into (1) and simplifying (see Appendix B), we obtain

$$I_{\mathbf{s}_a}(R) = E \frac{2-t}{2+t}, \quad t := -2 + \eta^{-1}. \quad (13)$$

Towards simplifying $|I_{\mathbf{s}_a}(R)| = E|2-t|/(2+t)$, we note that [see (2)]

$$2-t = 4 - R^K - R^{-K} \geq 0 \iff R^{2K} - 4R^K + 1 \leq 0.$$

The above is a quadratic equation in $x = R^K$ with the two roots $x_0 = (2 + \sqrt{3})^{\pm 1}$. Hence, for $R \geq R_0 = (2 + \sqrt{3})^{1/K}$, $2-t \leq 0$ such that $|2-t| = t-2$ and

$$|I_{\mathbf{s}_a}(R)| = E \frac{t-2}{t+2} = E(-4\eta + 1).$$

The lower bound in (★) follows as

$$d_{\min}^2 \geq 2E - 2|I_{\mathbf{s}_a}(R)| = 8E\eta,$$

which holds for even $K \geq 2$ and $R > 1$ sufficiently large. ■

Appendix A (Complex logarithm identities)

We first consider the identity in (8), which holds if and only if

$$\text{Arg}(1 - R^{-1}(\phi^*)^k \zeta^\ell) - \text{Arg}(1 - R^{-1}\phi^k \zeta^{-\ell}) \in (-\pi, \pi].$$

Indeed, as $R \rightarrow \infty$, $1 - R^{-1}(\phi^*)^k \zeta^\ell \rightarrow 1$ and $1 - R^{-1}\phi^k \zeta^{-\ell} \rightarrow 1$ so that

$$\text{Arg}(1 - R^{-1}(\phi^*)^k \zeta^\ell) - \text{Arg}(1 - R^{-1}\phi^k \zeta^{-\ell}) \rightarrow 0.$$

Thus the desired condition holds for sufficiently large $R > 1$.

We now consider the identity in the first line of (10), which holds if and only if

$$\sum_{k=0}^{K-1} s_k \text{Arg}(G_k(\zeta^\ell)) \in (-\pi, \pi].$$

In our case, $G_k(\zeta^\ell) \rightarrow 1$ as $R \rightarrow \infty$ so that $\text{Arg}(G_k(\zeta^\ell)) \rightarrow 0$. Hence $s_k \text{Arg}(G_k(\zeta^\ell)) \rightarrow 0$ and the above sum condition holds, at least for sufficiently large $R > 1$.

Appendix B (Simplifying $I_{\mathbf{s}_a}(R)$)

For even $K \geq 2$, define $\mathcal{K}_{\text{even}} := \{k \in [K] : k \text{ even}\}$ and $\mathcal{K}_{\text{odd}} := \{k \in [K] : k \text{ odd}\}$. Then, substituting $\mathbf{s}_a = [1, -1, \dots, 1, -1]^T$ into (3), we have

$$\begin{aligned} P_{\mathbf{s}_a}(\zeta^\ell) &= \prod_{k \in \mathcal{K}_{\text{even}}} B_k(\zeta^\ell) \prod_{k \in \mathcal{K}_{\text{odd}}} B_k(\zeta^\ell)^{-1} = \prod_{k \in \mathcal{K}_{\text{even}}} \frac{\zeta^\ell - R\phi^k}{\zeta^\ell - R^{-1}\phi^k} \prod_{k \in \mathcal{K}_{\text{odd}}} \frac{\zeta^\ell - R^{-1}\phi^k}{\zeta^\ell - R\phi^k} \\ &\stackrel{(d)}{=} \frac{(\zeta^{\ell N} - R^N)(\zeta^{\ell N} + R^{-N})}{(\zeta^{\ell N} - R^{-N})(\zeta^{\ell N} + R^N)}, \end{aligned}$$

where $N := K/2$ and in (d) we applied the identities

$$\prod_{k \in \mathcal{K}_{\text{even}}} (\zeta^\ell - R^{\pm 1}\phi^k) = (\zeta^{\ell N} - R^{\pm N}) \quad \text{and} \quad \prod_{k \in \mathcal{K}_{\text{odd}}} (\zeta^\ell - R^{\pm 1}\phi^k) = (\zeta^{\ell N} + R^{\pm N}).$$

Defining $\theta_\ell := \pi\ell/(K+1)$ and $\gamma := R^N - R^{-N}$, $P_{\mathbf{s}_a}(\zeta^\ell)$ further simplifies to

$$P_{\mathbf{s}_a}(\zeta^\ell) = \frac{(\zeta^{\ell N} - R^N)(\zeta^{\ell N} + R^{-N})}{(\zeta^{\ell N} - R^{-N})(\zeta^{\ell N} + R^N)} = \frac{\zeta^{\ell N} - \zeta^{-\ell N} - \gamma}{\zeta^{\ell N} - \zeta^{-\ell N} + \gamma} = \frac{2j \sin(K\theta_\ell) - \gamma}{2j \sin(K\theta_\ell) + \gamma}.$$

In particular, note that because $\sin(\cdot)$ is odd, $P_{\mathbf{s}_a}(\zeta^{K+1-\ell}) = P_{\mathbf{s}_a}^*(\zeta^\ell)$ for all $\ell \in \{1, \dots, K\}$. On the other hand, $A_H(\zeta^\ell)$ is even such that, for all $\ell \in \{1, \dots, K\}$, $A_H(\zeta^{K+1-\ell}) = A_H(\zeta^\ell)$ [see (2)]. Hence, for $\mathbf{s}_a = [1, -1, \dots, 1, -1]^T$, the terms in (1) come in conjugate pairs as

$$\begin{aligned} I_{\mathbf{s}_a}(R) &= \frac{1}{K+1} \sum_{\ell=0}^K A_H(\zeta^\ell) P_{\mathbf{s}_a}(\zeta^\ell) = \frac{1}{K+1} \left[A_H(1) P_{\mathbf{s}_a}(1) + \sum_{\ell=1}^K A_H(\zeta^\ell) P_{\mathbf{s}_a}(\zeta^\ell) \right] \\ &= \frac{1}{K+1} \left[E(2\eta - 1) + 2 \sum_{\ell=1}^N A_H(\zeta^\ell) \Re(P_{\mathbf{s}_a}(\zeta^\ell)) \right]. \end{aligned}$$

We have

$$\Re(P_{\mathbf{s}_a}(\zeta^\ell)) = \frac{4 \sin^2(\theta_\ell) - \gamma^2}{4 \sin^2(\theta_\ell) + \gamma^2} = \frac{a_\ell - t}{a_\ell + t}, \quad a_\ell := 4 \sin^2(\theta_\ell), \quad t := \gamma^2.$$

Further, $\eta = 1/(2+t)$, and $a_\ell = 2 - 2 \cos(2\theta_\ell)$ so that

$$A_H(\zeta^\ell) = E[1 - 2\eta \cos(2\theta_\ell)] = E \left[1 - \frac{2 \cos(2\theta_\ell)}{2+t} \right] = E \frac{2 - 2 \cos(2\theta_\ell) + t}{2+t} = E \frac{a_\ell + t}{2+t}.$$

The product $A_H(\zeta^\ell) \Re(P_{\mathbf{s}_a}(\zeta^\ell))$ thus has the form

$$A_H(\zeta^\ell) \Re(P_{\mathbf{s}_a}(\zeta^\ell)) = E \frac{a_\ell + t}{2+t} \cdot \frac{a_\ell - t}{a_\ell + t} = E \frac{a_\ell - t}{2+t}.$$

We can now simplify $I_{\mathbf{s}_a}(R)$ on the previous page to obtain (13) as

$$\begin{aligned}
I_{\mathbf{s}_a}(R) &= \frac{1}{K+1} \left[E(2\eta - 1) + 2 \sum_{\ell=1}^N E \frac{a_\ell - t}{2+t} \right] \\
&= \frac{E}{K+1} \left[-\frac{t}{2+t} + 2 \sum_{\ell=1}^N \frac{a_\ell - t}{2+t} \right] \\
&= \frac{E}{K+1} \cdot \frac{-t + 2 \sum_{\ell=1}^N a_\ell - Kt}{2+t} \\
&\stackrel{(e)}{=} \frac{E}{K+1} \cdot \frac{2(K+1) - (K+1)t}{2+t} \\
&= E \frac{2-t}{2+t},
\end{aligned}$$

where (e) follows from $\sum_{\ell=1}^N a_\ell = K+1$.

References

- [1] P. Huggins, A. Şahin, and E. Erkip, “On the distance properties of Huffman sequences,” *IEEE Commun. Lett.*, pp. 1–5, 2026.